

3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- a. Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

OLS一定通过平均点 $(\bar{x}, \bar{y})$

$$INSURANCE = 6.855 + 3.880 \times 59.3$$

$$\Rightarrow INSURANCE = 236.939$$

b. How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.

① 点估计: 每增加\$1,000收入, 保险会增加多少 $\Rightarrow$ 变化量

点估计的 $b_2 = 3,880$

我们估计, 每增加 1,000 美元的家庭收入, 持有的保险平均金额增加 3,880 美元

② 区间估计 95% CI

$$CI = b_2 \pm t_c \cdot Se(b_2)$$

$$\bullet b_2 = 3,880$$

$$\bullet Se(b_2) = 0.112$$

$$\bullet df = N - 2 = 20 - 2 = 18$$

$$.95 = P(t_{18; 0.025} < \frac{b_2 - 3,880}{Se(b_2)} < t_{18; 0.975})$$

$$= P(-2.101 \times 0.112 + 3,880 < b_2 < 2.101 \times 0.112 + 3,880,$$

$$= P(3,644 < b_2 < 4,115)$$

③ Explain

我们有 95% 信心, 一个家庭每增加 1,000 美元的收入, 他购买人壽保险的平均金额就会增加 \$3,644 到 \$4,115 美元。

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .

共变異數

Step 1: 計算點估計 ($\hat{y}_0$)

$$\begin{aligned}\hat{y} &= b_1 + b_2 X_0 \\ &= 6.855 + 3.880 \times 100,000 \\ &= 394,855\end{aligned}$$

Step 2: Calculate the Variance of the Estimate

$$\widehat{\text{Var}}(\hat{y}_0) = \widehat{\text{var}}(b_1) + x^2 \widehat{\text{var}}(b_2) + 2x_0 \widehat{\text{cov}}(b_1, b_2)$$

$$\begin{aligned}\widehat{\text{Var}}(\hat{y}_0) &= (7.383)^2 + 100^2 (0.112)^2 + 2 \times 100 \times (-0.746) \\ &= 30.74\end{aligned}$$

$$se = \sqrt{30.74} \approx 5.545$$

Step 3: 建立 99% CI

$$\bullet df = N - 2 = 18$$

$$\bullet \text{臨界值 } t_c: \alpha = 0.01, t_{18; 0.995} \approx 2.878$$

$$99\% = P(t_{18; 0.005} < \frac{y - 394.855}{5.545} < t_{18; 0.995})$$

$$= P(-2.878 \times 5.545 + 394.855 < y < 2.878 \times 5.545 + 394.855)$$

$$= P(378.90 < y < 410.81)$$

⇒ 我們有 99% 的信心，當家庭收入為 \$100,000，其平均保險金額會落在 \$378,897 到 \$410,813 美元之間。

- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $INSURANCE = \beta_1 + \beta_2 INCOME + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

每增加\$1,000收入,保險額增加\$5,000? ($\beta_2 = 5$?)

1. 建立假設 又又尾

$$H_0: \beta_2 = 5 \text{ (宣稱屬實)}$$

$$H_1: \beta_2 \neq 5 \text{ (宣稱不屬實)}$$

2. 計算檢定量(t)

$$t = \frac{b_2 - \beta_{2,H_0}}{se(b_2)} = \frac{3.880 - 5}{0.112} = -10$$

3. 拒絕域

$$\alpha = 0.05$$

$$df = 18$$

$$t_{\alpha/2} = 0.025 \quad t_c \approx 2.101$$

$$\text{拒絕域 } t \leq -2.101 \text{ 或 } t \geq 2.101$$

4. 結論

$-10 < -2.101$, 拒絕 H_0 , 不支持管理層說:

- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

保險額隨收入同等增加 (斜率=1)

1. 建立假設 單尾, 方向性

$$H_0: \beta_2 = 1$$

$$H_1: \beta_2 > 1 \text{ (右尾)}$$

2. 計算檢定量(t)

$$t = \frac{b_2 - \beta_{2,H_0}}{se(b_2)} = \frac{3.880 - 1}{0.112} \approx 25.71$$

3. 拒絕域

$$\alpha = 1\% \quad df = 18$$

$$t_{18, 0.99} = 2.552$$

$$\text{拒絕域 } t > 2.552$$

4. 結論

$$25.71 > 2.552$$

拒絕 H_0 。支持 H_1 ，當收入增加時，
保險額增加金額超過同等金額。

3.23 The data file *collegetown* contains data on 500 single-family houses sold in Baton Rouge, Louisiana, during 2009–2013. The data include sale price in \$1000 units, PRICE, and total interior area in hundreds of square feet, SQFT.

- a. Using the quadratic regression model, $PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$, test the hypothesis that the marginal effect on expected house price of increasing the size of a 2000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000. Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

1. 邊際效應

$$PRICE = \alpha_1 + \alpha_2 SQFT^2 + e$$

$$ME = \frac{d(PRICE)}{d(SQFT)} = 2\alpha_2 SQFT$$

$$SQFT = 20$$

$$ME = 40\alpha_2$$

2. 建立假設¹

$$H_0: 40\alpha_2 \leq 13$$

$$H_1: 40\alpha_2 > 13 \text{ (右尾)}$$

3. 計算 t 統計量

```
#安裝並載入讀取資料的套件
library(POE5Rdata)
#讀取數據
data("collegetown")
#幫助文件 (Help) 變數定義
?collegetown
#建立二次項模型
mod1 <- lm(price~I(sqft^2), data = collegetown)
#提取結果摘要
smod1<-summary(mod1)
smod1
```

```
Call:
lm(formula = price ~ I(sqft^2), data = collegetown)

Residuals:
    Min       1Q   Median       3Q      Max
-383.67  -48.39   -7.50   38.75  469.70

Coefficients:
             $\alpha_2$                  $Se(\alpha_2)$ 
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 93.565854    6.072226   15.41  <2e-16 ***
I(sqft^2)    0.184519 ✓ 0.005256 ✓ 35.11  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92.08 on 498 degrees of freedom
Multiple R-squared:  0.7122,    Adjusted R-squared:  0.7117
F-statistic: 1233 on 1 and 498 DF,  p-value: < 2.2e-16
```

$$t = \frac{40 \cdot \alpha_2 - 13}{40 \cdot Se(\alpha_2)}$$
$$= \frac{40 \times 0.1845 - 13}{40 \times 0.005256} = -26.728$$

F 拒絕域

$$df = 500 - 2 = 498$$

$$\alpha = 0.05$$

$$\text{臨界值 } t_{498, 0.05} \approx 1.645$$

5. 結論

$t = -26.73$ 遠小於臨界值 1.645

⇒ 無法拒絕 H_0

沒有證據支持 ME 大於 13,000 的說法

- b. Using the quadratic regression model in part (a), test the hypothesis that the marginal effect on expected house price of increasing the size of a 4000 square foot house by 100 square feet is less than or equal to \$13000 against the alternative that the marginal effect will be greater than \$13000.

Use the 5% level of significance. Clearly state the test statistic used, the rejection region, and the test p -value. What do you conclude?

1. data

$$\alpha_2 = 0.184519$$

$$Se(\alpha_2) = 0.005256$$

$$SQFT = 40$$

$$\text{目標值 } \$13,000 \Rightarrow 13$$

2. 計算新 $\widehat{ME}$

$$ME = 2 \cdot \alpha_2 \cdot SQFT$$

$$\widehat{ME} = 2 \cdot 0.184519 \cdot 40 = 14.76152$$

$\Rightarrow$ 在 4,000 呎時, 每增加 100 呎, 房所預計增加 \$14,761.52

3. 計算新的 $Se(ME)$

$$Se(\widehat{ME}) = 80 \cdot Se(\alpha_2) = 80 \times 0.005256 = 0.42048$$

4. 假設檢定

1. 建立假設

$$H_0: ME \leq 13$$

$$H_1: ME > 13 \text{ (右尾)}$$

2. 計算t統計量

$$t = \frac{14.76152 - 13}{0.42048} \approx 4.189$$

3. 判定拒絕域或P值 Δ

臨界值 (t_c): $\alpha = 0.05$ $df = 498$ $t_c \approx 1.645$
拒絕域 $t \geq 1.645$

$\Rightarrow 4.189 > 1.645 \Rightarrow$ 拒絕 H_0

5. 結論

對於1棟4000平方英尺的房子來說
增加100平方英尺對預期價格的迴降影響
在統計上顯著大於13,000美元。

```
> #part b
> tstar<-(80*a2-13)/(80*seaz
> tstar
I(sqft^2)
4.189467
> alpha<-0.05
> df=df.residual(mod1)
> tc<-qt(1-alpha,df)
> tc
[1] 1.647919
```

- c. Using the quadratic regression model in part (a), estimate the expected price $E(PRICE|SQFT) = \alpha_1 + \alpha_2 SQFT^2$ for a house of 2000 square feet. Construct a 95% interval estimate of the expected price. Describe your interval estimate to a general audience.

1. 點估計

期望價格

$$E(PRICE|SQFT = 2000/100) = 167.3735$$

$$SQFT = 20$$

$$PRICE = \alpha_1 + \alpha_2 \cdot SQFT^2$$

$$93.56585 + 0.184519 \times (20)^2 \widehat{PRICE} = \alpha_1 + \alpha_2 \cdot (20)^2$$

2. 單位換算

$$167.3735 \times 1000 = 167,373.5$$

3. 信心區間 95%

$$(155,594.1, 179,152.8)$$

我們有95%信心，房子的真實平均價格會落在15.5萬到17.9萬美元之間

4. 結論

信心區間指如果重新抽樣100次
大約會有95個會抓到真正的市場平均價。

```
> #part c
> x=2000/100 #SQFT
> a1
(Intercept)
  93.56585
> a2
I(sqft^2)
  0.184519
> #截距項的變異數
> vara1<-vcov(mod1)[1,1]
> vara2<-vcov(mod1)[2,2]
> #a1截距與斜率之間的共變異數
> cova1a2<-vcov(mod1)[1,2]
> L<-a1+a2*x^2
> L
(Intercept)
  167.3735

> #線性組合變異數
> varL=vara1+x^2*vara2+2*x*cova1a2
> seL<-sqrt(varL)
> alpha<-0.05
> tcr<-qt(1-alpha/2,df)
> lowbl<-L-tcr*seL
> upbl<-L+tcr*seL
> lowbl
(Intercept)
  155.5941
> upbl
(Intercept)
  179.1528
```

- d. Locate houses in the sample with 2000 square feet of living area. Calculate the sample mean (average) of their selling prices. Is the sample average of the selling price for houses with $SQFT = 20$ compatible with the result in part (c)? Explain.

求特定樣本平均值 并用歸模型預測值

```
> #part d
> collegetown$price[which(collegetown$sqft==20)]
[1] 138 169 183
> p20<-mean(collegetown$price[which(collegetown$sqft==20)])
> p20
[1] 163.3333
> |
```